

WINTER HAVEN, FL, USA

WMO: 747931

Lat: 28.062N

Lon: 81.754W

Elev: 45

StdP: 100.79

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 12876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.8	6.1	-6.2	2.3	9.0	-2.9	3.0	11.5	9.5	17.3	8.6	16.5	3.4	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	34.7	24.1	33.9	24.1	33.1	24.0	26.1	30.6	25.7	30.2	25.5	29.9	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	27.5	24.7	19.8	27.3	24.2	19.3	27.1	81.0	30.5	79.3	30.4	78.0	30.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.4	7.5	6.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	0.1	36.3	2.2	0.7	-1.5	36.9	-2.7	37.3	-3.9	37.7	-5.5	38.2	
			WB	-1.9	26.8	2.2	0.6	-3.4	27.2	-4.7	27.5	-5.9	27.8	-7.4	28.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	16.1	18.1	19.9	23.0	25.8	27.8	28.4	28.5	27.7	24.6	20.3	18.2
	DBStd	5.30	4.52	4.38	3.58	2.64	2.08	1.48	1.31	1.24	1.25	3.02	3.58	4.40
	HDD10.0	14	8	2	0	0	0	0	0	0	0	0	0	3
	HDD18.3	263	95	53	27	3	0	0	0	0	0	0	22	58
	CDD10.0	4848	197	230	308	392	491	534	571	574	530	453	309	257
	CDD18.3	2056	26	48	77	145	233	284	313	316	280	198	82	54
	CDH23.3	18094	139	302	529	1162	2095	2683	3110	3186	2565	1570	479	275
Wind	CDH26.7	6877	8	52	112	369	889	1155	1303	1313	1016	532	90	37
	WSAvg	3.1	3.2	3.2	3.6	3.6	3.4	2.8	2.4	2.6	2.9	3.2	3.2	3.1
	PrecAvg	1322	71	60	91	71	79	220	197	212	165	63	40	70
	PrecMax	1825	154	112	198	165	223	346	299	296	372	158	98	313
	PrecMin	996	16	14	8	10	31	87	89	116	64	3	2	16
	PrecStd	219	40	33	65	40	54	81	56	51	90	44	28	83
	DB	27.7	29.7	30.3	32.5	35.0	35.9	35.5	35.3	34.5	33.3	30.9	29.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	20.5	21.0	20.7	21.8	22.0	24.2	24.5	25.0	24.6	24.2	23.3	22.2	
	DB	26.3	28.3	29.1	31.2	33.5	34.3	34.3	34.2	33.3	32.1	28.9	27.7	
	MCWB	19.5	20.1	20.0	21.1	21.9	23.7	24.5	24.7	24.1	23.9	22.1	21.2	
	DB	25.0	27.0	27.9	30.1	32.5	33.3	33.5	33.5	32.6	31.0	27.5	26.4	
	MCWB	18.7	19.4	19.3	20.5	21.7	23.7	24.5	24.7	24.0	23.3	20.6	20.6	
	DB	23.2	25.3	26.7	28.9	31.3	32.4	32.5	32.6	31.8	29.7	26.2	24.9	
	MCWB	18.2	18.8	18.7	19.8	21.5	23.7	24.5	24.7	24.0	22.1	19.7	19.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	21.9	22.6	22.9	23.9	24.7	26.2	26.7	26.8	26.3	25.7	24.2	23.4	
	MCDB	25.7	26.9	26.5	28.3	29.2	30.8	30.9	31.4	30.4	29.8	28.2	26.9	
	WB	21.0	21.6	21.9	22.9	24.0	25.5	25.8	26.2	25.7	25.1	23.4	22.5	
	MCDB	24.0	25.4	25.8	27.8	28.8	30.4	30.4	30.6	29.9	29.3	27.0	25.8	
	WB	20.2	20.7	21.2	22.2	23.5	25.1	25.6	25.8	25.4	24.6	22.5	21.7	
	MCDB	23.2	24.5	25.0	27.0	28.4	30.0	30.1	30.2	29.5	28.5	25.6	24.4	
	WB	19.1	20.0	20.3	21.6	23.0	24.7	25.2	25.5	25.0	24.0	21.4	20.9	
Mean Daily Temperature Range	MCDB	22.2	23.9	24.3	26.4	27.8	29.3	29.8	29.9	29.0	27.6	24.5	23.6	
	MDBR	11.0	11.2	11.3	11.3	10.8	9.4	8.8	8.7	8.7	9.8	10.4	10.3	
	5% DB	11.5	11.7	11.6	11.7	11.8	10.6	9.7	9.7	9.4	9.6	10.3	10.3	
	MCWBR	5.4	4.7	4.1	3.7	3.0	2.8	2.6	2.6	2.6	2.9	4.3	4.7	
	5% WB	9.8	10.2	9.9	10.3	9.9	9.1	8.7	8.6	8.3	8.5	8.9	9.1	
	MCWBR	5.8	5.0	4.7	4.2	3.2	2.9	2.9	2.9	2.7	2.8	4.3	5.0	
	taub	0.339	0.351	0.368	0.385	0.422	0.450	0.472	0.457	0.421	0.394	0.357	0.342	
Clear-Sky Solar Irradiance	taud	2.512	2.489	2.434	2.364	2.308	2.305	2.241	2.305	2.390	2.431	2.486	2.530	
	Ebn at Noon	905	921	923	912	873	843	823	836	859	864	877	885	
	Edh at Noon	91	101	113	125	132	132	140	131	117	106	93	86	
	RadAvg	3.51	4.18	5.25	6.20	6.46	5.75	5.65	5.43	4.94	4.55	3.75	3.22	
All-Sky Solar Radiation	RadStd	0.25	0.25	0.39	0.30	0.42	0.50	0.36	0.36	0.27	0.30	0.19	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-4	-45	+158	+112

Nomenclature: See separate page